

Wenhao Peng

RF Acoustic Wave Devices

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Biography

Wenhao Peng is an RF acoustic wave engineer at Skyworks Solutions, working across device modeling, filter design, and high-volume manufacturing of bulk acoustic wave (BAW) resonators and filters. His work spans thin-film piezoelectric and ferroelectric materials, including aluminum nitride, scandium-doped aluminum nitride, and barium strontium titanate, applied to intrinsically switchable filters and millimeter-wave FBARs. He developed the first equivalent-circuit model deriving ferroelectric BAW resonator behavior from ferroelectric thermodynamics, connecting dielectric, elastic, and piezoelectric effects to electromechanical coupling across multilayer stacks, and demonstrated the first FBARs operating above 50 GHz using periodically poled AlN/ScAlN/AlN trilayers. He received the Ph.D. degree in electrical and computer engineering from the University of Michigan, Ann Arbor, in 2026, and has published in IEEE TMTT, IEEE MWTL, IEEE MWCL, and Applied Physics Letters.

Selected Contributions

- mm-Wave scaling** First FBAR demonstrated above 50 GHz: a 56 GHz periodically poled AlN/ScAlN/AlN trilayer resonator $k_t^2 = 1.7\%$ and $Q_m = 53$, extending BAW technology roughly five times beyond the prior state of the art below 10 GHz. A second-generation 36 GHz device, optimized for electrode metal and layer stack, raised coupling and quality factor to $k_t^2 = 2.9\%$ and $Q_m = 102$. Sole first author on both (IMS 2024, IMS 2025).
- Physics-based modeling** First equivalent-circuit model of ferroelectric-driven BAW resonators derived from ferroelectric thermodynamics rather than empirical fitting, capturing the combined dielectric, elastic, and piezoelectric contributions to coupling. First author, IEEE TMTT 2022.
- Coupling framework** First analytical framework relating acoustic field distribution to k_t^2 via modal analysis and series-parallel resonance separation, replacing iterative full-wave sweeps with a closed-form route to layer stack design and enabling the 36 GHz coupling improvement above.
- Switchless frontends** Co-developed the first switchless quad-band filter bank built on ferroelectric BST FBARs.
- Materials** Co-demonstrated controlled ferroelectric polarization switching in ultrawide-bandgap AlN/ScAlN multilayers (Applied Physics Letters 2023), the materials advance that made the periodically poled mm-wave FBARs above possible.

Experience

- Feb 2026–Present **RF Acoustic Staff Electrical Engineer**, *Skyworks Solutions, Inc.*, Irvine, CA.
- Serve as technical bridge across device modeling, filter design, and high-volume manufacturing, ensuring RF designs translate reliably into mass-production outcomes.
 - Contribute to resonator stack design and optimization for higher-frequency cellular bands, balancing performance, tunability, and manufacturability requirements.
 - Developing methods to accelerate mask-level acoustic response simulation, targeting shorter design-iteration cycles and improved predictive accuracy ahead of fabrication (ongoing R&D).
 - Apply equivalent-circuit and multiphysics modeling to guide design decisions and troubleshoot performance across the design-to-production pipeline, working with modeling, process, and production teams to align design targets with fab capability and yield.
- Sep 2019–Dec 2025 **Graduate Student Research Assistant**, *University of Michigan*, Ann Arbor, MI.
Advisor: Prof. Amir Mortazawi
- 2025 *mm-Wave acoustic resonators based on periodically poled III-nitride thin films*
- Demonstrated the first mm-wave FBARs using AlN/ScAlN/AlN periodically poled trilayers, reaching 56 GHz ($k_t^2 = 1.7\%$, $Q = 53$) and subsequently 36 GHz with optimized coupling and quality factor ($k_t^2 = 2.9\%$, $Q = 102$).
 - Derived an analytical framework relating acoustic field distribution to k_t^2 through modal analysis and series-parallel resonance separation, cutting layer stack design from iterative optimizations based on admittance responses to closed-form evaluation.
 - Ran the full design-to-characterization workflow: Mason equivalent-circuit models (ADS), multiphysics simulation (COMSOL), electrode metal and layer stack optimization, and analysis of intrinsic material Q limits.
 - Advanced nanofabrication and polarization-switching processes, including lithography, dry/wet/plasma etching, physical vapor deposition, and pulsed-field poling with current-response analysis.
 - Performed mm-wave characterization to 67 GHz using on-wafer de-embedding structures, GSG probes, and VNA, with test structures validated in ADS Momentum and Ansys HFSS.
- Reconfigurable RF frontends based on intrinsically switchable BST acoustic resonators*
- Built the first equivalent-circuit model of ferroelectric-driven BAW resonators grounded in ferroelectric thermodynamics, capturing the underlying dielectric, elastic, and piezoelectric physics rather than fitting measured response.
 - Contributed to the first switchless quad-band BST FBAR filter bank and the first intrinsically switchable balanced ferroelectric FBAR filter at 2 GHz.
 - Contributed to the first RF acoustic phase inversion switch based on poled-ferroelectric stacked crystal filters.
 - Designed resonators and filters with Mason and mBVD models in ADS, using the image parameter method for filter synthesis.
 - Executed full cleanroom fabrication (oxidation, lithography, physical vapor deposition, wet/dry etching) and characterized devices under DC bias via GSG probes and VNA.

Education

- 2026 **Ph.D., Electrical and Computer Engineering**, *University of Michigan*, Ann Arbor, MI.
Dissertation: *Layered Ferroelectric Millimeter-Wave BAW Resonators: Modeling and Validation based on On-Wafer Characterization*. Advisor: Prof. Amir Mortazawi.
- 2024 **M.S.E., Mechanical Engineering**, *University of Michigan*, Ann Arbor, MI.
Dynamics and vibrations

- 2019 **M.S., Electrical and Computer Engineering**, *University of Michigan*, Ann Arbor, MI.
Integrated circuits and VLSI
- 2018 **B.S.E., Electrical Engineering**, *University of Michigan*, Ann Arbor, MI.
Summa Cum Laude
- 2018 **B.S.E., Electrical and Computer Engineering**, *Shanghai Jiao Tong University*, Shanghai, China.
Dual degree with U-M

Technical Skills

Devices & materials	BAW/FBAR resonator and filter design; thin-film piezoelectrics and ferroelectrics (AlN, ScAlN, BST); periodically poled III-nitride multilayers; MEMS; RF frontend architectures
Modeling	Mason and mBVD equivalent circuits; thermodynamic modeling of ferroelectric nonlinearity; modal analysis; image-parameter filter synthesis; multiphysics FEM (COMSOL); EM co-simulation (ADS Momentum, Ansys HFSS); structural vibration analysis
Fabrication	Cleanroom process integration; photolithography; physical vapor deposition; wet/dry/plasma etching; thermal oxidation; device processing compatible with MBE-grown films; pulsed-field poling
Measurement	On-wafer S-parameter characterization to 67 GHz; GSG probing; VNA calibration and de-embedding; DC-bias-dependent RF characterization; material parameter extraction
Tools	ADS, COMSOL Multiphysics, Ansys HFSS, MATLAB, Python, Cadence, LLM

Journal Publications

- 2025 H. Desai, **W. Peng**, and A. Mortazawi, "Single-Pole Single-Throw RF Acoustic Phase Inversion Switch Leveraging Poled Ferroelectrics," *IEEE Trans. Microw. Theory Techn.*, vol. 73, no. 1, pp. 6–13, 2025.
- 2023 D. Wang, P. Wang, S. Mondal, J. Liu, M. Hu, M. He, S. Nam, **W. Peng**, S. Yang, D. Wang, Y. Xiao, Y. Wu, A. Mortazawi, and Z. Mi, "Controlled Ferroelectric Switching in Ultrawide Bandgap AlN/ScAlN Multilayers," *Appl. Phys. Lett.*, vol. 123, no. 10, p. 103506, 2023.
- 2023 S. Nam, **W. Peng**, P. Wang, D. Wang, Z. Mi, and A. Mortazawi, "A mm-Wave Trilayer AlN/ScAlN/AlN Higher Order Mode FBAR," *IEEE Microw. Wireless Technol. Lett.*, vol. 33, no. 6, pp. 803–806, 2023.
- 2022 **W. Peng**, M. Z. Koohi, S. Nam, and A. Mortazawi, "Phenomenological Circuit Modeling of Ferroelectric-Driven Bulk Acoustic Wave Resonators," *IEEE Trans. Microw. Theory Techn.*, vol. 70, no. 1, pp. 919–925, 2022.
- 2021 S. Nam, M. Z. Koohi, **W. Peng**, and A. Mortazawi, "A Switchless Quad Band Filter Bank Based on Ferroelectric BST FBARs," *IEEE Microw. Wireless Compon. Lett.*, vol. 31, no. 6, pp. 662–665, 2021.

Conference Publications

- 2025 **W. Peng**, S. Nam, D. Wang, Z. Mi, and A. Mortazawi, “A 36 GHz Trilayer AlN/ScAlN/AlN Periodically Poled FBAR,” *IEEE/MTT-S International Microwave Symposium (IMS)*, San Francisco, CA, 2025, pp. 786–789. *(presented by author)*
- 2024 **W. Peng**, S. Nam, D. Wang, Z. Mi, and A. Mortazawi, “A 56 GHz Trilayer AlN/ScAlN/AlN Periodically Poled FBAR,” *IEEE/MTT-S International Microwave Symposium (IMS)*, Washington, DC, 2024, pp. 150–153. *(presented by author)*
- 2021 **W. Peng**, M. Z. Koohi, S. Nam, and A. Mortazawi, “Physics Based Modeling of Electrostriction Based BAW Resonators,” *IEEE MTT-S International Microwave Symposium (IMS)*, Atlanta, GA, 2021, pp. 214–217. *(presented by author)*
- 2020 M. Z. Koohi, **W. Peng**, and A. Mortazawi, “An Intrinsically Switchable Balanced Ferroelectric FBAR Filter at 2 GHz,” *IEEE/MTT-S International Microwave Symposium (IMS)*, 2020, pp. 131–134.
- 2018 Y. Dai, **W. Peng**, Y. Wang, L.-X. Chuo, K. Suri, H. Zheng, D. Wentzloff, and H.-S. Kim, “Implementation and Evaluation of Bi-Directional WiFi Back-Channel Communication,” *IEEE 29th Annual International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*, 2018, pp. 1–7.

Peer Review

40 reviews **IEEE:** J. Microelectromechanical Systems (7), Microwave and Wireless Technology Letters (6), Electron Device Letters (4), J. Microwaves (1). **APS:** Physical Review Letters (5), Physical Review Applied (2), Physical Review Materials (2), Physical Review B (1). **Other:** Micromachines (7), Photonics (2), Microwave (2), Progress in Quantum Electronics (1).

Selected Graduate Coursework

Acoustics & mechanics Wave Propagation in Elastic Solids; Advanced Vibrations of Structures; Advanced Engineering Acoustics; Theory of Solid Continua; Mathematical Methods in Mechanical Engineering

RF & circuits Advanced Solid State Microwave Circuits; Electromagnetics Theory; Microwave Circuits; Introduction to MEMS; Monolithic Amplifier Circuits; VLSI Design I/II